

Supplementary Material for:

The Potential Impact of a Four-Point Shot: A Statistical Simulation Approach

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This document serves as the Supplementary Material for the main paper, "The Potential Impact of a Four-Point Shot: A Statistical Simulation Approach." The primary study investigates how a four-point shot might alter the competitive landscape and strategic dynamics of the National Basketball Association (NBA), a topic often discussed in response to the strategic monotony of the "three-point era."

As detailed in the main paper's abstract, our methodology is a probabilistic statistical simulation that models each possession as a hierarchical series of probabilistic choices. These choices are governed by Dirichlet distributions, which are calibrated using optimized parameters (ω and β) to realistically reproduce the 2023-24 NBA season.

The purpose of this supplement is to provide a detailed, granular breakdown of this simulation model and the complete datasets used for its calibration and execution. The following sections present the full hierarchical structure of the model (Figures S1-S5) and the corresponding data tables. These tables include the α proportions for every Dirichlet distribution and the success-rate probabilities (e.g., FG%, Opponent FG%) for each event. This data is provided for the baseline "Reproduced Season" model (Figures S1-S4) as well as for the two 4-point shot experimental scenarios, Case 1 (Direct Conversion) and Case 2 (Player Tendency Model), which are detailed in Figure S5.

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Table of Acronyms

Table S1: List of acronyms used in the Supplementary Material.

Acronym	Description
FGA	Field Goal Attempts
FGM (or FG)	Field Goals Made
FTA	Free Throw Attempts
MP	Minutes Played
2PA	2-point Attempt
3PA	3-point Attempt
4PA	4-point Attempt
FTD	Free Throw Drawn
TOV	Turnover
ORB	Offensive Rebound Percentage
DRB	Defensive Rebound Percentage
RAA	Restricted Area Attempt
PZA	Paint Zone Attempt
MRA	Mid-Range Attempt
2PAnd1	2-point And-1
RAP	Restricted Area Percentage (Team FG%)
PZP	Paint Zone Percentage (Team FG%)
MRP	Mid-Range Percentage (Team FG%)
ORAP	Opponent Restricted Area Percentage Allowed (Opp FG%)
OPZP	Opponent Paint Zone Percentage Allowed (Opp FG%)
OMRP	Opponent Mid-Range Percentage Allowed (Opp FG%)
FTP	Free Throw Percentage
COA	Corner-3 Attempt
ABA	Above the break Attempt
DEEPA	Deep-3 Attempt
3PAnd1	3-point And-1
COP	Corner-3 Percentage (Team FG%)
ABP	Above the break Percentage (Team FG%)
DEEPP	Deep-3 Percentage (Team FG%)
OCOP	Opponent Corner-3 Percentage Allowed (Opp FG%)
OABP	Opponent Above the break Percentage Allowed (Opp FG%)
4PP	4-point Percentage (Team FG%)
2SFD	2-point Shot Foul Drawn
3SFD	3-point Shot Foul Drawn
NSFD	Non shooting Foul Drawn

1 Possession and Pace Formulae

This section details the standard NBA statistical formulae used to calculate game possessions and season pace, which are fundamental to our simulation's calibration and analysis. All acronyms used in these formulae are defined in the Table of Acronyms (S1).

1.1 Game Possession Formula

The number of possessions for a single game ($\text{Poss}_{\text{game}}$) is estimated using the following formula, which averages the possession counts calculated from both the team's (Tm) and the opponent's (Opp) box score statistics:

$$\text{Poss}_{\text{game}} = 0.5 \times \left[\begin{aligned} & \left(\text{Tm FGA} + 0.4 \times \text{Tm FTA} \right. \\ & \quad \left. - 1.07 \times \frac{\text{Tm ORB}}{\text{Tm ORB} + \text{Opp DRB}} \times (\text{Tm FGA} - \text{Tm FGM}) + \text{Tm TOV} \right) \\ & + \left(\text{Opp FGA} + 0.4 \times \text{Opp FTA} \right. \\ & \quad \left. - 1.07 \times \frac{\text{Opp ORB}}{\text{Opp ORB} + \text{Tm DRB}} \times (\text{Opp FGA} - \text{Opp FGM}) + \text{Opp TOV} \right) \end{aligned} \right] \quad (\text{S1})$$

1.2 Pace Formula

Pace is a measure of the number of possessions a team averages per 48 minutes of play. It is calculated using the season totals for possessions (Tm Poss and Opp Poss) and total team minutes played (Tm MP).

$$\text{PACE} = 48 \times \left(\frac{\text{Tm Poss} + \text{Opp Poss}}{2 \times (\text{Tm MP}/5)} \right) \quad (\text{S2})$$

In S2, Tm Poss and Opp Poss represent the season totals of the two main components within the brackets of S1, respectively. The term Tm MP/5 gives the total number of game minutes (including overtime) played by the team.

2 Supplementary Figures

2.1 Figure S1: Possession

This figure illustrates the top-level structure of our possession model. A single possession is modeled as a hierarchical process beginning with a primary outcome determined by a Dirichlet distribution: $\text{Dirichlet}(\omega \cdot (2\text{PA}, 3\text{PA}, \text{FTD}, \text{TOV}))$. The ω parameter serves as a weighting factor. The '2PA', '3PA', and 'FTD' branches lead to the sub-models detailed in Figures S2, S3, and S4, respectively. A 'TOV' immediately ends the possession with $n = 0$ points. The model also accounts for offensive rebounds; if a field goal is missed (except in the case of a TOV), the possession is replayed if an offensive rebound occurs, which is modeled as a binomial trial: $\text{Bin}(1, \frac{\text{ORB}}{\text{Opp DRB}}) = 1$.

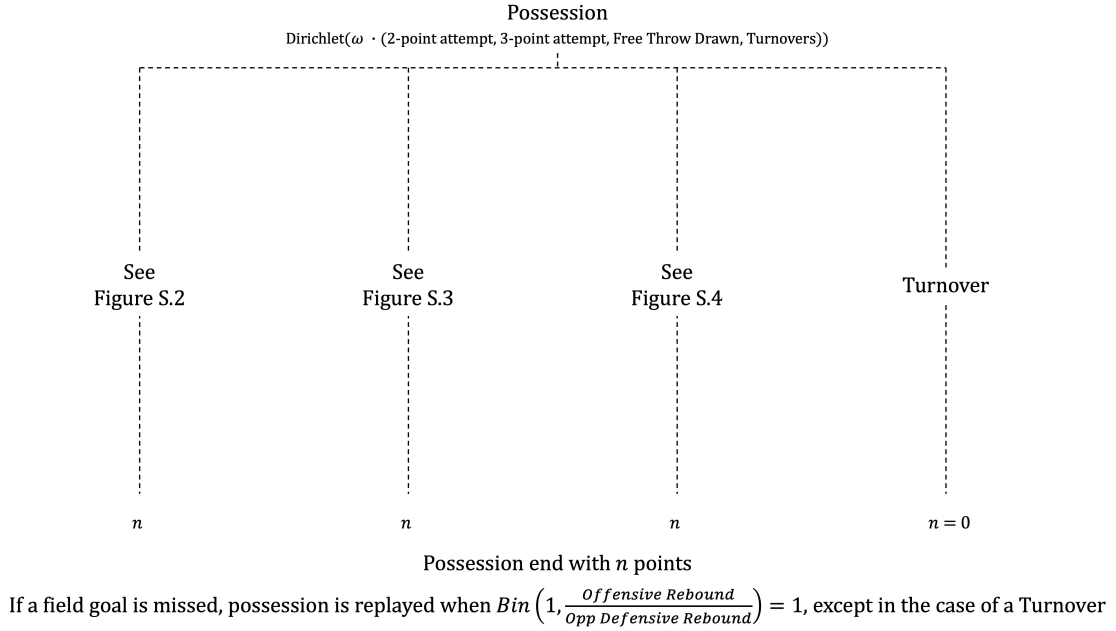


Figure S1: Top-level possession model (Base Model).

2.2 Figure S2: 2-point

If a possession results in a '2PA' from Figure S1, this sub-model is triggered. The 2PA is further classified into one of four types using a Dirichlet distribution: 'RAA', 'PZA', 'MRA', or '2PAnd1'. For standard field goals (RAA, PZA, MRA), the success probability (p) is calculated as a weighted average of the team's offensive field goal percentage in that zone (RAP, PZP, MRP) and the opponent's defensive field goal percentage allowed in that zone (ORAP, OPZP, OMRP), governed by the formula $p = \beta \cdot \text{Team FG\%} + (1 - \beta) \cdot \text{Opp FG\%}$. The resulting points n are determined by a binomial trial $n = 2 \times \text{Bin}(1, p)$, yielding 2 points for a success and 0 for a failure. An '2PAnd1' event results in a successful 2-point basket plus a free throw, yielding $n = 2 + \text{Bin}(1, p_{FT})$ points. The free throw probability p_{FT} is simply the team's FTP.

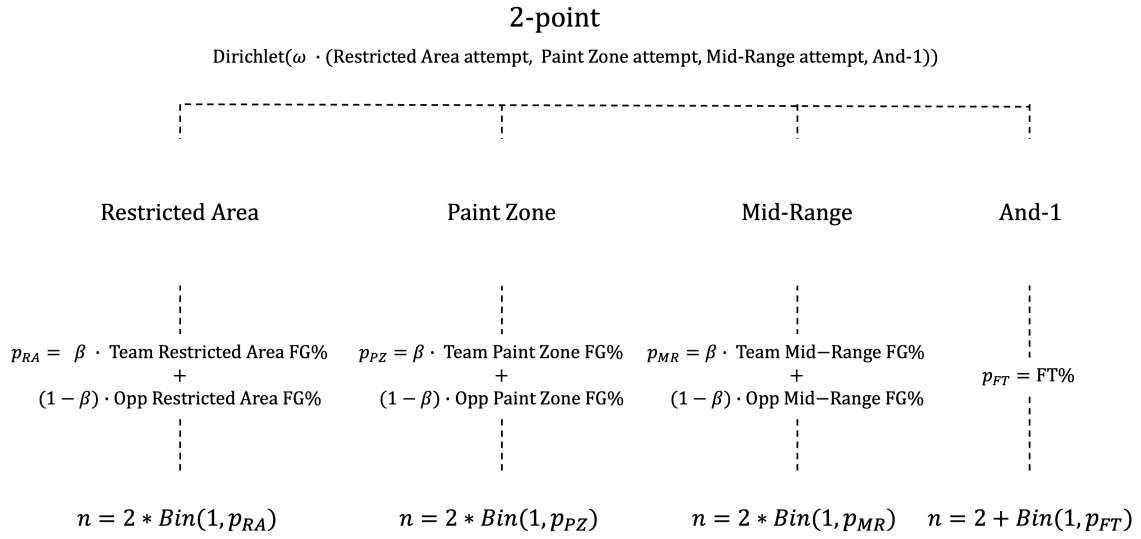


Figure S2: 2-point attempt (2PA) sub-model.

2.3 Figure S3: 3-point

This figure details the '3PA' sub-model. This attempt type is classified into four categories via a Dirichlet distribution: 'COA', 'ABA', 'DEEPA', or '3PAnd1'. The success probabilities for 'COA' (p_{CO}) and 'ABA' (p_{AB}) are calculated using the same β -weighted formula as in the 2-point model, using COP/ABP and OCOP/OABP data respectively. However, the 'DEEPA' shot probability (p_{Deep}) is assumed to be unaffected by defense and is set directly to the team's DEEPP. The points scored n are determined by $n = 3 \times \text{Bin}(1, p)$, resulting in 3 points for a make and 0 for a miss. An '3PAnd1' (a 4-point play) results in $n = 3 + \text{Bin}(1, p_{FT})$ points.

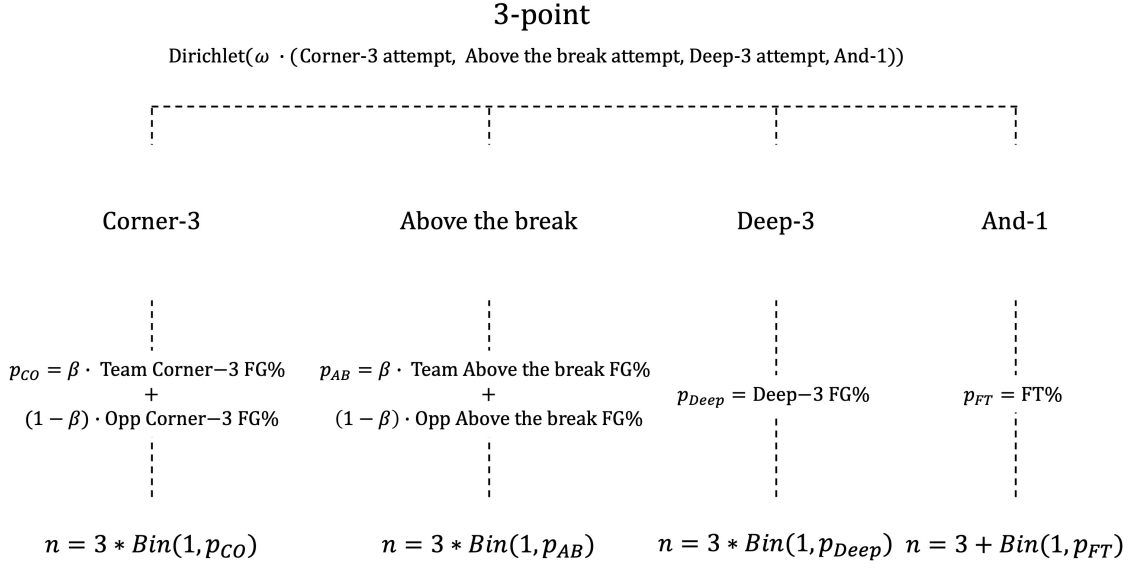


Figure S3: 3-point attempt (3PA) sub-model (Base Model).

2.4 Figure S4: Free Throw

The 'FTD' sub-model governs possessions ending in a foul. These events are drawn from a Dirichlet distribution with three categories: '2SFD', '3SFD', and 'NSFD'. A '2SFD' or a 'NSFD' (such as a bonus situation) results in two free throws, with the points scored n determined by $n = \text{Bin}(2, p_{FT})$. A '3SFD' results in three free throws, with $n = \text{Bin}(3, p_{FT})$. The success probability p_{FT} is the team's FTP, which is provided in S5 and S8.

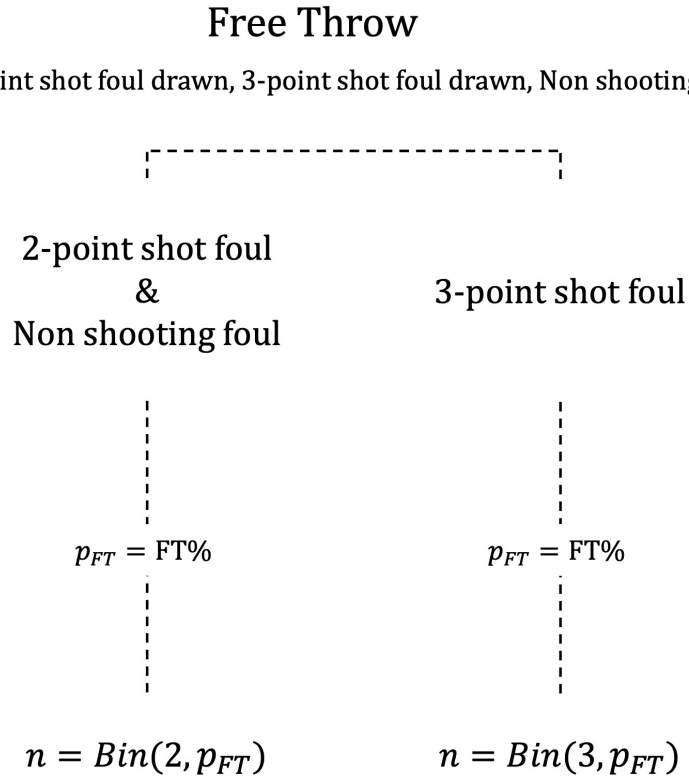


Figure S4: Free Throw Drawn (FTD) sub-model.

2.5 Figure S5: Possession (4-point Shot Model)

This figure presents the modified possession model used for our 4-point shot simulations. It adapts the top-level model from Figure S1 by introducing a '4PA' branch. This structure serves as the foundation for two distinct simulation scenarios: ****Case 1 (Direct Conversion)**** and ****Case 2 (Player Tendency Model)****. In this model, a 4PA's success is determined by $n = 4 \times \text{Bin}(1, p_{4P})$, where p_{4P} is the team's 4PP. The '3PA' branch is modified, containing a nested Dirichlet distribution for 'COA', 'ABA', and '3PAand1' attempts, as depicted in the diagram.

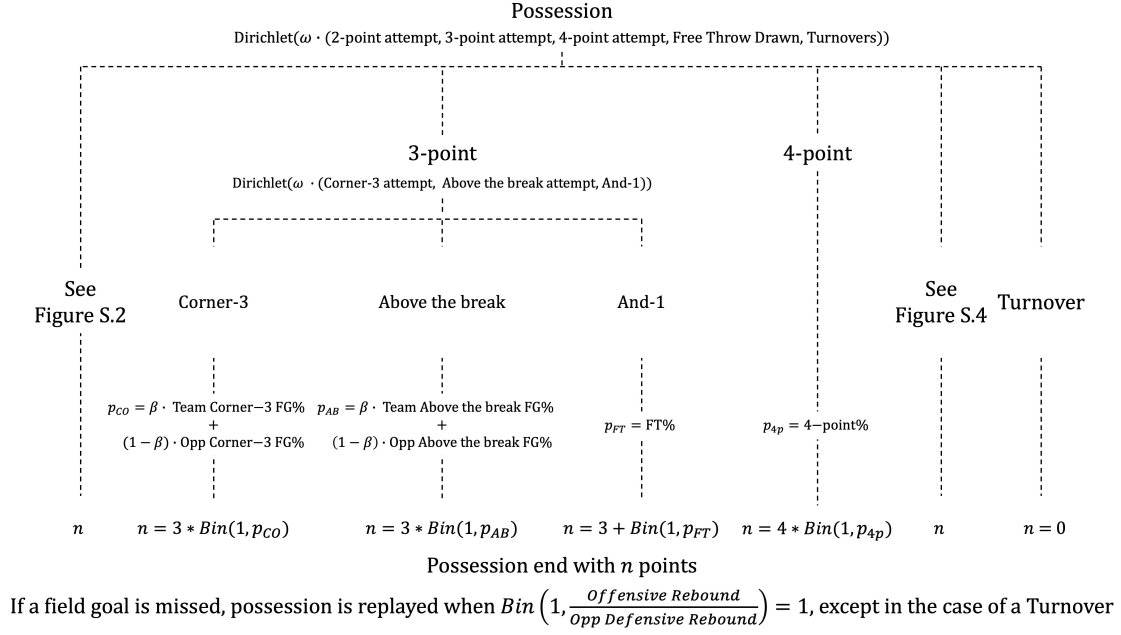


Figure S5: Alternative possession model incorporating a 4-point attempt (4PA).

3 Supplementary Tables

3.1 Data for Figure S1 (Possession)

The proportions for the top-level Dirichlet distribution are provided in S2. The data used to calculate the offensive rebound probability is provided in S3.

Table S2: Possession Outcome Proportions (2PA, 3PA, FTD, TOV) for the **Reproduced Season** simulation.

Team	2PA	3PA	FTD	TOV
Atlanta Hawks	0.436	0.358	0.108	0.098
Boston Celtics	0.415	0.408	0.104	0.098
Brooklyn Nets	0.421	0.378	0.102	0.099
Charlotte Hornets	0.454	0.361	0.101	0.121
Chicago Bulls	0.479	0.340	0.107	0.095
Cleveland Cavaliers	0.445	0.386	0.103	0.105
Dallas Mavericks	0.395	0.426	0.095	0.094
Denver Nuggets	0.465	0.301	0.099	0.106
Detroit Pistons	0.445	0.352	0.112	0.118
Golden State Warriors	0.405	0.417	0.090	0.106
Houston Rockets	0.439	0.380	0.106	0.094
Indiana Pacers	0.480	0.337	0.103	0.102
LA Clippers	0.465	0.350	0.108	0.102
Los Angeles Lakers	0.493	0.321	0.108	0.100
Memphis Grizzlies	0.481	0.340	0.104	0.110
Miami Heat	0.450	0.363	0.109	0.103
Milwaukee Bucks	0.453	0.365	0.112	0.095
Minnesota Timberwolves	0.484	0.328	0.110	0.109
New Orleans Pelicans	0.470	0.341	0.109	0.102
New York Knicks	0.468	0.347	0.114	0.095
Oklahoma City Thunder	0.473	0.342	0.112	0.099
Orlando Magic	0.480	0.327	0.115	0.107
Philadelphia 76ers	0.456	0.344	0.120	0.097
Phoenix Suns	0.483	0.328	0.115	0.104
Portland Trail Blazers	0.438	0.369	0.105	0.118
Sacramento Kings	0.424	0.394	0.097	0.100
San Antonio Spurs	0.477	0.338	0.098	0.113
Toronto Raptors	0.490	0.320	0.103	0.106
Utah Jazz	0.462	0.352	0.112	0.114
Washington Wizards	0.471	0.338	0.101	0.108

Table S3: Team Offensive (ORB) and Defensive (DRB) Rebound Percentages for the **Reproduced Season** simulation.

Team	ORB (%)	DRB (%)
Atlanta Hawks	12.5	32.2
Boston Celtics	10.7	35.6
Brooklyn Nets	11.4	32.6
Charlotte Hornets	9.3	31.0
Chicago Bulls	11.2	32.6
Cleveland Cavaliers	9.8	33.4
Dallas Mavericks	9.7	33.2
Denver Nuggets	10.7	33.7
Detroit Pistons	10.5	32.8
Golden State Warriors	12.1	34.6
Houston Rockets	11.5	34.0
Indiana Pacers	10.1	31.4
LA Clippers	10.0	32.9
Los Angeles Lakers	8.2	34.9
Memphis Grizzlies	10.9	31.7
Miami Heat	9.3	33.0
Milwaukee Bucks	9.4	34.8
Minnesota Timberwolves	9.4	34.2
New Orleans Pelicans	10.4	33.6
New York Knicks	12.7	32.5
Oklahoma City Thunder	8.8	33.2
Orlando Magic	10.5	31.8
Philadelphia 76ers	11.0	31.9
Phoenix Suns	10.1	33.9
Portland Trail Blazers	12.6	30.1
Sacramento Kings	10.8	33.2
San Antonio Spurs	10.4	33.9
Toronto Raptors	10.9	31.8
Utah Jazz	12.2	33.2
Washington Wizards	9.2	31.9

3.2 Data for Figure S2 (2-point)

S4 provides the α proportions for the 2PA Dirichlet distribution. The subsequent tables (S5 and S6) provide the team and opponent field goal percentages used to calculate the success probabilities (p).

Table S4: 2PA Proportions (RAA, PZA, MRA, 2PAnd1) for the **Reproduced Season** simulation.

Team	RAA	PZA	MRA	2PAnd1
Atlanta Hawks	0.474	0.327	0.157	0.041
Boston Celtics	0.484	0.293	0.183	0.040
Brooklyn Nets	0.480	0.310	0.171	0.039
Charlotte Hornets	0.479	0.292	0.190	0.039
Chicago Bulls	0.458	0.261	0.244	0.037
Cleveland Cavaliers	0.524	0.307	0.133	0.036
Dallas Mavericks	0.447	0.291	0.213	0.049
Denver Nuggets	0.478	0.279	0.205	0.038
Detroit Pistons	0.449	0.319	0.194	0.037
Golden State Warriors	0.417	0.339	0.206	0.039
Houston Rockets	0.490	0.292	0.173	0.045
Indiana Pacers	0.494	0.297	0.159	0.049
LA Clippers	0.445	0.321	0.196	0.039
Los Angeles Lakers	0.482	0.293	0.177	0.049
Memphis Grizzlies	0.487	0.356	0.120	0.037
Miami Heat	0.396	0.383	0.184	0.037
Milwaukee Bucks	0.481	0.248	0.226	0.045
Minnesota Timberwolves	0.472	0.330	0.150	0.048
New Orleans Pelicans	0.469	0.320	0.172	0.039
New York Knicks	0.463	0.324	0.166	0.047
Oklahoma City Thunder	0.466	0.304	0.189	0.042
Orlando Magic	0.519	0.267	0.169	0.044
Philadelphia 76ers	0.476	0.304	0.173	0.047
Phoenix Suns	0.413	0.298	0.247	0.042
Portland Trail Blazers	0.469	0.311	0.185	0.035
Sacramento Kings	0.434	0.360	0.164	0.041
San Antonio Spurs	0.493	0.297	0.171	0.039
Toronto Raptors	0.481	0.319	0.156	0.044
Utah Jazz	0.437	0.401	0.120	0.042
Washington Wizards	0.487	0.323	0.143	0.046

Table S5: Team FG% by Zone (RAP, PZP, MRP) and FTP.

Team	RAP (%)	PZP (%)	MRP (%)	FTP (%)
Atlanta Hawks	62.3	44.0	46.8	79.7
Boston Celtics	70.3	45.4	43.1	80.7
Brooklyn Nets	64.5	40.7	38.6	75.6
Charlotte Hornets	65.2	41.7	38.2	78.6
Chicago Bulls	62.2	46.4	43.7	79.1
Cleveland Cavaliers	65.8	43.8	46.2	76.5
Dallas Mavericks	70.7	45.9	42.8	75.8
Denver Nuggets	65.1	48.8	45.5	76.2
Detroit Pistons	64.4	42.8	42.6	78.5
Golden State Warriors	70.3	44.1	41.0	78.0
Houston Rockets	64.0	41.6	41.2	77.3
Indiana Pacers	68.5	47.3	50.5	78.2
LA Clippers	67.0	48.2	41.4	82.5
Los Angeles Lakers	72.4	41.8	38.6	78.2
Memphis Grizzlies	60.6	40.8	36.2	76.4
Miami Heat	63.4	48.6	38.0	81.8
Milwaukee Bucks	70.8	42.8	45.0	77.4
Minnesota Timberwolves	67.5	43.3	39.1	77.7
New Orleans Pelicans	65.9	44.1	44.5	77.1
New York Knicks	63.1	45.9	39.2	78.0
Oklahoma City Thunder	67.5	46.8	46.0	82.5
Orlando Magic	67.2	42.1	37.6	75.9
Philadelphia 76ers	63.1	42.0	41.6	82.6
Phoenix Suns	67.5	47.4	47.4	80.8
Portland Trail Blazers	60.4	37.7	41.2	79.1
Sacramento Kings	68.7	48.2	39.9	74.5
San Antonio Spurs	66.0	43.5	37.2	78.2
Toronto Raptors	67.5	41.2	40.6	75.6
Utah Jazz	68.7	44.4	35.8	83.0
Washington Wizards	69.7	39.9	37.7	76.4

Table S6: Opponent FG% Allowed by Zone (ORAP, OPZP, OMRP).

Team	ORAP (%)	OPZP (%)	OMRP (%)
Atlanta Hawks	67.2	45.0	42.7
Boston Celtics	64.0	42.9	42.0
Brooklyn Nets	65.7	39.9	43.1
Charlotte Hornets	67.6	46.2	42.1
Chicago Bulls	66.1	43.6	42.1
Cleveland Cavaliers	63.0	41.7	40.9
Dallas Mavericks	67.8	44.5	40.2
Denver Nuggets	63.8	41.7	39.7
Detroit Pistons	67.9	45.1	44.9
Golden State Warriors	67.3	48.9	39.2
Houston Rockets	65.9	44.8	43.0
Indiana Pacers	65.3	47.7	44.3
LA Clippers	67.4	41.1	42.3
Los Angeles Lakers	68.1	41.9	42.3
Memphis Grizzlies	63.5	42.7	42.9
Miami Heat	67.5	47.3	43.3
Milwaukee Bucks	66.0	46.2	42.3
Minnesota Timberwolves	63.2	41.2	39.8
New Orleans Pelicans	67.9	43.3	41.0
New York Knicks	63.5	43.5	41.8
Oklahoma City Thunder	62.1	43.6	39.7
Orlando Magic	65.8	44.8	42.9
Philadelphia 76ers	66.7	39.8	40.0
Phoenix Suns	65.2	42.1	38.5
Portland Trail Blazers	70.5	45.9	42.1
Sacramento Kings	66.5	45.4	41.9
San Antonio Spurs	65.9	44.5	43.7
Toronto Raptors	69.3	44.8	40.7
Utah Jazz	67.4	45.7	42.2
Washington Wizards	71.0	42.8	44.7

3.3 Data for Figure S3 (3-point)

S7 provides the α proportions for the 3PA Dirichlet distribution. The subsequent tables (S8 and S9) provide the team and opponent field goal percentages used to calculate the success probabilities.

Table S7: 3PA Proportions (COA, ABA, DEEPA, 3PAnd1) for the **Reproduced Season** simulation.

Team	COA	ABA	DEEPA	3PAnd1
Atlanta Hawks	0.269	0.684	0.042	0.005
Boston Celtics	0.216	0.763	0.019	0.001
Brooklyn Nets	0.268	0.707	0.020	0.005
Charlotte Hornets	0.222	0.736	0.040	0.002
Chicago Bulls	0.282	0.697	0.019	0.001
Cleveland Cavaliers	0.241	0.720	0.035	0.004
Dallas Mavericks	0.279	0.694	0.023	0.004
Denver Nuggets	0.245	0.733	0.022	0.000
Detroit Pistons	0.278	0.697	0.024	0.002
Golden State Warriors	0.169	0.792	0.035	0.000
Houston Rockets	0.178	0.798	0.023	0.001
Indiana Pacers	0.241	0.734	0.024	0.001
LA Clippers	0.261	0.722	0.012	0.004
Los Angeles Lakers	0.260	0.709	0.030	0.002
Memphis Grizzlies	0.253	0.723	0.023	0.002
Miami Heat	0.294	0.688	0.015	0.003
Milwaukee Bucks	0.257	0.706	0.034	0.003
Minnesota Timberwolves	0.291	0.695	0.013	0.001
New Orleans Pelicans	0.270	0.703	0.025	0.002
New York Knicks	0.282	0.705	0.010	0.000
Oklahoma City Thunder	0.256	0.735	0.009	0.001
Orlando Magic	0.268	0.708	0.023	0.002
Philadelphia 76ers	0.222	0.755	0.021	0.001
Phoenix Suns	0.253	0.729	0.017	0.002
Portland Trail Blazers	0.248	0.725	0.024	0.002
Sacramento Kings	0.251	0.736	0.012	0.001
San Antonio Spurs	0.240	0.743	0.014	0.003
Toronto Raptors	0.261	0.720	0.019	0.000
Utah Jazz	0.285	0.698	0.014	0.003
Washington Wizards	0.226	0.727	0.044	0.003

Table S8: Team 3-point FG% (COP, ABP, DEEPP) and FTP.

Team	COP (%)	ABP (%)	DEEPP (%)	FTP (%)
Atlanta Hawks	38.0	36.0	23.7	79.7
Boston Celtics	43.0	37.7	23.2	80.7
Brooklyn Nets	41.1	34.6	19.4	75.6
Charlotte Hornets	36.1	35.6	20.2	78.6
Chicago Bulls	38.5	34.9	22.0	79.1
Cleveland Cavaliers	37.1	36.8	23.6	76.5
Dallas Mavericks	36.3	37.4	18.2	75.8
Denver Nuggets	38.4	37.6	10.7	76.2
Detroit Pistons	36.2	34.7	12.5	78.5
Golden State Warriors	38.6	38.0	21.7	78.0
Houston Rockets	39.4	34.6	15.9	77.3
Indiana Pacers	40.9	36.5	26.0	78.2
LA Clippers	41.0	37.2	8.8	82.5
Los Angeles Lakers	38.7	37.6	30.8	78.2
Memphis Grizzlies	34.9	34.8	16.7	76.4
Miami Heat	39.3	36.2	12.2	81.8
Milwaukee Bucks	41.2	36.0	30.0	77.4
Minnesota Timberwolves	40.8	38.1	13.9	77.7
New Orleans Pelicans	42.6	37.0	22.4	77.1
New York Knicks	40.7	35.5	29.0	78.0
Oklahoma City Thunder	41.4	38.1	20.0	82.5
Orlando Magic	36.7	34.9	8.3	75.9
Philadelphia 76ers	39.7	35.4	22.0	82.6
Phoenix Suns	40.6	37.4	33.3	80.8
Portland Trail Blazers	36.6	34.0	25.0	79.1
Sacramento Kings	38.6	36.0	13.2	74.5
San Antonio Spurs	37.8	34.0	14.6	78.2
Toronto Raptors	37.7	33.9	20.8	75.6
Utah Jazz	38.9	34.3	14.0	83.0
Washington Wizards	38.4	33.9	21.1	76.4

Table S9: Opponent 3-point FG% Allowed (OCOP, OABP).

Team	OCOP (%)	OABP (%)
Atlanta Hawks	40.6	37.8
Boston Celtics	35.2	35.6
Brooklyn Nets	41.8	36.1
Charlotte Hornets	40.0	37.1
Chicago Bulls	41.6	35.3
Cleveland Cavaliers	39.4	36.5
Dallas Mavericks	38.6	36.2
Denver Nuggets	37.5	35.3
Detroit Pistons	39.2	36.5
Golden State Warriors	38.0	35.4
Houston Rockets	36.9	34.3
Indiana Pacers	39.8	35.8
LA Clippers	37.3	36.1
Los Angeles Lakers	39.6	37.2
Memphis Grizzlies	41.2	37.0
Miami Heat	37.7	34.6
Milwaukee Bucks	36.8	35.5
Minnesota Timberwolves	38.6	34.6
New Orleans Pelicans	37.9	34.0
New York Knicks	39.7	35.5
Oklahoma City Thunder	36.9	36.0
Orlando Magic	36.5	35.9
Philadelphia 76ers	38.4	34.7
Phoenix Suns	35.9	36.8
Portland Trail Blazers	40.3	33.6
Sacramento Kings	40.4	38.2
San Antonio Spurs	37.6	37.5
Toronto Raptors	40.2	37.0
Utah Jazz	45.2	37.9
Washington Wizards	40.3	35.0

3.4 Data for Figure S4 (Free Throw)

S10 provides the α proportions for the foul type Dirichlet distribution.

Table S10: FTD Proportions (2SFD, 3SFD, NSFD) for the **Reproduced Season** simulation.

Team	2SFD	3SFD	NSFD
Atlanta Hawks	0.786	0.058	0.156
Boston Celtics	0.809	0.033	0.158
Brooklyn Nets	0.793	0.046	0.161
Charlotte Hornets	0.839	0.031	0.131
Chicago Bulls	0.822	0.012	0.166
Cleveland Cavaliers	0.793	0.034	0.173
Dallas Mavericks	0.810	0.033	0.157
Denver Nuggets	0.777	0.022	0.202
Detroit Pistons	0.838	0.031	0.131
Golden State Warriors	0.816	0.045	0.139
Houston Rockets	0.838	0.014	0.148
Indiana Pacers	0.854	0.012	0.134
LA Clippers	0.780	0.066	0.154
Los Angeles Lakers	0.805	0.024	0.170
Memphis Grizzlies	0.789	0.032	0.179
Miami Heat	0.802	0.034	0.164
Milwaukee Bucks	0.763	0.047	0.190
Minnesota Timberwolves	0.805	0.013	0.183
New Orleans Pelicans	0.838	0.028	0.134
New York Knicks	0.809	0.040	0.151
Oklahoma City Thunder	0.829	0.017	0.154
Orlando Magic	0.819	0.016	0.165
Philadelphia 76ers	0.846	0.018	0.136
Phoenix Suns	0.784	0.031	0.185
Portland Trail Blazers	0.834	0.028	0.138
Sacramento Kings	0.831	0.019	0.150
San Antonio Spurs	0.834	0.039	0.127
Toronto Raptors	0.836	0.017	0.147
Utah Jazz	0.789	0.042	0.169
Washington Wizards	0.847	0.023	0.130

3.5 Data for Figure S5 (4-point Shot Model)

Case 1: Direct Conversion (> 29 ft)

In this scenario, the '4PA' is created by directly reclassifying the 'DEEPA' attempts (defined as shots > 29 ft) from the original 3-point model (Figure S3). The 'DEEPA' category is removed from the '3PA' sub-model and becomes its own top-level category ('4PA' in S11). Consequently, the '3PA' branch in this case (S12) consists only of 'COA', 'ABA', and '3PAnd1' attempts.

S11 provides the top-level possession proportions for Case 1. The 4-point success probability (p_{4P}) for this case is the team's DEEPP from S8. S12 provides the proportions for the nested 3PA model specific to Case 1.

Table S11: Case 1 Top-Level Possession Proportions (2PA, 3PA, 4PA, TOV, FTD).

Team	2PA	3PA	4PA	TOV	FTD
Atlanta Hawks	0.463	0.305	0.014	0.114	0.105
Boston Celtics	0.423	0.369	0.007	0.106	0.095
Brooklyn Nets	0.462	0.317	0.007	0.116	0.098
Charlotte Hornets	0.479	0.294	0.013	0.124	0.090
Chicago Bulls	0.508	0.278	0.005	0.108	0.101
Cleveland Cavaliers	0.452	0.318	0.012	0.122	0.097
Dallas Mavericks	0.439	0.338	0.008	0.110	0.106
Denver Nuggets	0.513	0.272	0.007	0.113	0.096
Detroit Pistons	0.492	0.269	0.007	0.132	0.100
Golden State Warriors	0.452	0.321	0.012	0.123	0.092
Houston Rockets	0.471	0.304	0.007	0.110	0.108
Indiana Pacers	0.491	0.294	0.008	0.110	0.098
LA Clippers	0.480	0.294	0.004	0.118	0.104
Los Angeles Lakers	0.490	0.266	0.008	0.122	0.114
Memphis Grizzlies	0.441	0.322	0.013	0.132	0.092
Miami Heat	0.473	0.302	0.005	0.115	0.105
Milwaukee Bucks	0.442	0.322	0.012	0.113	0.111
Minnesota Timberwolves	0.468	0.289	0.003	0.127	0.112
New Orleans Pelicans	0.487	0.282	0.007	0.116	0.108
New York Knicks	0.465	0.312	0.003	0.116	0.103
Oklahoma City Thunder	0.485	0.298	0.003	0.112	0.102
Orlando Magic	0.476	0.271	0.006	0.130	0.116
Philadelphia 76ers	0.491	0.286	0.006	0.105	0.111
Phoenix Suns	0.472	0.282	0.005	0.131	0.109
Portland Trail Blazers	0.488	0.280	0.007	0.132	0.094
Sacramento Kings	0.448	0.337	0.004	0.114	0.097
San Antonio Spurs	0.466	0.308	0.004	0.130	0.092
Toronto Raptors	0.491	0.281	0.006	0.121	0.101
Utah Jazz	0.454	0.306	0.004	0.133	0.102
Washington Wizards	0.479	0.291	0.014	0.120	0.095

Table S12: Case 1 Nested 3PA Proportions (COA, ABA, 3PAnd1).

Team	COA	ABA	3PAnd1
Atlanta Hawks	0.281	0.714	0.005
Boston Celtics	0.220	0.778	0.002
Brooklyn Nets	0.274	0.721	0.005
Charlotte Hornets	0.232	0.767	0.002
Chicago Bulls	0.288	0.710	0.002
Cleveland Cavaliers	0.250	0.746	0.004
Dallas Mavericks	0.285	0.711	0.004
Denver Nuggets	0.250	0.749	0.000
Detroit Pistons	0.284	0.714	0.002
Golden State Warriors	0.175	0.821	0.000
Houston Rockets	0.182	0.817	0.001
Indiana Pacers	0.247	0.752	0.001
LA Clippers	0.265	0.731	0.004
Los Angeles Lakers	0.268	0.731	0.002
Memphis Grizzlies	0.259	0.740	0.002
Miami Heat	0.299	0.698	0.003
Milwaukee Bucks	0.266	0.731	0.003
Minnesota Timberwolves	0.295	0.704	0.001
New Orleans Pelicans	0.277	0.721	0.002
New York Knicks	0.285	0.712	0.000
Oklahoma City Thunder	0.258	0.741	0.001
Orlando Magic	0.274	0.724	0.002
Philadelphia 76ers	0.227	0.771	0.001
Phoenix Suns	0.257	0.741	0.002
Portland Trail Blazers	0.254	0.743	0.002
Sacramento Kings	0.254	0.745	0.001
San Antonio Spurs	0.244	0.753	0.003
Toronto Raptors	0.266	0.733	0.000
Utah Jazz	0.289	0.708	0.003
Washington Wizards	0.236	0.761	0.003

Case 2: Player Tendency Model (> 25 ft Above-the-Break)

This scenario models a more significant strategic shift. The '4PA' category ('4PA' in S13) is created by reclassifying all 'ABA' 3PA attempts taken from beyond 25 ft. The '3PA' branch ('3PA' in S13) now represents the remaining 3-point shots. The nested 3PA model for this case (S14) reflects the new proportions of 'COA' and the modified, shorter-range 'ABA' shots.

S13 provides the top-level possession proportions for Case 2. S14 provides the proportions for the nested 3PA model. Tables S15 and S16 provide the team and opponent field goal percentages, respectively, used to calculate success probabilities for the 3PA and 4PA shots in this scenario.

Table S13: Case 2 Top-Level Possession Proportions (2PA, 3PA, 4PA, TOV, FTD).

Team	2PA	3PA	4PA	TOV	FTD
Atlanta Hawks	0.463	0.119	0.200	0.114	0.105
Boston Celtics	0.423	0.108	0.268	0.106	0.095
Brooklyn Nets	0.462	0.125	0.198	0.116	0.098
Charlotte Hornets	0.479	0.095	0.218	0.124	0.090
Chicago Bulls	0.508	0.118	0.165	0.108	0.101
Cleveland Cavaliers	0.452	0.114	0.216	0.122	0.097
Dallas Mavericks	0.439	0.131	0.215	0.110	0.106
Denver Nuggets	0.513	0.114	0.165	0.113	0.096
Detroit Pistons	0.492	0.106	0.170	0.132	0.100
Golden State Warriors	0.452	0.083	0.250	0.123	0.092
Houston Rockets	0.471	0.085	0.226	0.110	0.108
Indiana Pacers	0.491	0.094	0.207	0.110	0.098
LA Clippers	0.480	0.102	0.196	0.118	0.104
Los Angeles Lakers	0.490	0.095	0.179	0.122	0.114
Memphis Grizzlies	0.441	0.122	0.208	0.132	0.092
Miami Heat	0.473	0.118	0.189	0.115	0.105
Milwaukee Bucks	0.442	0.110	0.224	0.113	0.111
Minnesota Timberwolves	0.468	0.136	0.156	0.127	0.112
New Orleans Pelicans	0.487	0.108	0.181	0.116	0.108
New York Knicks	0.465	0.131	0.184	0.116	0.103
Oklahoma City Thunder	0.485	0.117	0.184	0.112	0.102
Orlando Magic	0.476	0.096	0.182	0.130	0.116
Philadelphia 76ers	0.491	0.083	0.210	0.105	0.111
Phoenix Suns	0.472	0.100	0.187	0.131	0.109
Portland Trail Blazers	0.488	0.105	0.182	0.132	0.094
Sacramento Kings	0.448	0.117	0.224	0.114	0.097
San Antonio Spurs	0.466	0.110	0.202	0.130	0.092
Toronto Raptors	0.491	0.099	0.187	0.121	0.101
Utah Jazz	0.454	0.125	0.186	0.133	0.102
Washington Wizards	0.479	0.090	0.215	0.120	0.095

Table S14: Case 2 Nested 3PA Proportions (COA, ABA, 3PAnd1).

Team	COA	ABA	3PAnd1
Atlanta Hawks	0.674	0.313	0.013
Boston Celtics	0.720	0.273	0.007
Brooklyn Nets	0.674	0.313	0.012
Charlotte Hornets	0.667	0.328	0.005
Chicago Bulls	0.663	0.334	0.004
Cleveland Cavaliers	0.654	0.335	0.010
Dallas Mavericks	0.710	0.281	0.009
Denver Nuggets	0.589	0.410	0.001
Detroit Pistons	0.702	0.294	0.004
Golden State Warriors	0.615	0.370	0.015
Houston Rockets	0.626	0.371	0.004
Indiana Pacers	0.739	0.259	0.002
LA Clippers	0.746	0.241	0.012
Los Angeles Lakers	0.717	0.279	0.004
Memphis Grizzlies	0.661	0.334	0.004
Miami Heat	0.750	0.242	0.008
Milwaukee Bucks	0.732	0.260	0.008
Minnesota Timberwolves	0.618	0.379	0.002
New Orleans Pelicans	0.699	0.296	0.006
New York Knicks	0.668	0.325	0.007
Oklahoma City Thunder	0.651	0.347	0.002
Orlando Magic	0.752	0.244	0.004
Philadelphia 76ers	0.749	0.246	0.005
Phoenix Suns	0.704	0.291	0.005
Portland Trail Blazers	0.653	0.341	0.006
Sacramento Kings	0.720	0.278	0.002
San Antonio Spurs	0.666	0.326	0.007
Toronto Raptors	0.732	0.267	0.001
Utah Jazz	0.697	0.296	0.007
Washington Wizards	0.695	0.296	0.009

Table S15: Case 2 Team 4PP, 3P (COP, ABP), and FTP.

Team	COP (%)	ABP (%)	4PP (%)	FTP (%)
Atlanta Hawks	38.0	36.1	35.1	79.7
Boston Celtics	43.0	39.3	37.1	80.7
Brooklyn Nets	41.1	35.1	34.0	75.6
Charlotte Hornets	36.1	33.5	35.0	78.6
Chicago Bulls	38.5	34.7	34.5	79.1
Cleveland Cavaliers	37.1	34.7	36.4	76.5
Dallas Mavericks	36.3	40.0	36.2	75.8
Denver Nuggets	38.4	36.0	37.0	76.2
Detroit Pistons	36.2	30.5	34.6	78.5
Golden State Warriors	38.6	34.0	37.8	78.0
Houston Rockets	39.4	35.6	33.9	77.3
Indiana Pacers	40.9	34.0	36.4	78.2
LA Clippers	41.0	35.2	37.0	82.5
Los Angeles Lakers	38.7	37.8	37.3	78.2
Memphis Grizzlies	34.9	33.3	34.4	76.4
Miami Heat	39.3	38.0	35.4	81.8
Milwaukee Bucks	41.2	32.9	36.1	77.4
Minnesota Timberwolves	40.8	39.2	37.1	77.7
New Orleans Pelicans	42.6	37.9	36.2	77.1
New York Knicks	40.7	35.4	35.4	78.0
Oklahoma City Thunder	41.4	40.8	37.2	82.5
Orlando Magic	36.7	36.6	33.7	75.9
Philadelphia 76ers	39.7	35.8	35.0	82.6
Phoenix Suns	40.6	41.9	36.6	80.8
Portland Trail Blazers	36.6	36.8	33.0	79.1
Sacramento Kings	38.6	35.6	35.7	74.5
San Antonio Spurs	37.8	32.4	33.8	78.2
Toronto Raptors	37.7	29.4	34.2	75.6
Utah Jazz	38.9	28.9	34.9	83.0
Washington Wizards	38.4	28.3	33.9	76.4

Table S16: Case 2 Opponent 3P FG% Allowed (OCOP, OABP).

Team	OCOP (%)	OABP (%)
Atlanta Hawks	40.6	37.8
Boston Celtics	35.2	35.6
Brooklyn Nets	41.8	36.1
Charlotte Hornets	40.0	37.1
Chicago Bulls	41.6	35.3
Cleveland Cavaliers	39.4	36.5
Dallas Mavericks	38.6	36.2
Denver Nuggets	37.5	35.3
Detroit Pistons	39.2	36.5
Golden State Warriors	38.0	35.4
Houston Rockets	36.9	34.3
Indiana Pacers	39.8	35.8
LA Clippers	37.3	36.1
Los Angeles Lakers	39.6	37.2
Memphis Grizzlies	41.2	37.0
Miami Heat	37.7	34.6
Milwaukee Bucks	36.8	35.5
Minnesota Timberwolves	38.6	34.6
New Orleans Pelicans	37.9	34.0
New York Knicks	39.7	35.5
Oklahoma City Thunder	36.9	36.0
Orlando Magic	36.5	35.9
Philadelphia 76ers	38.4	34.7
Phoenix Suns	35.9	36.8
Portland Trail Blazers	40.3	33.6
Sacramento Kings	40.4	38.2
San Antonio Spurs	37.6	37.5
Toronto Raptors	40.2	37.0
Utah Jazz	45.2	37.9
Washington Wizards	40.3	35.0